

The Impact of Electrification on Labor Productivity: Evidence from the Early 20th Century United States

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Abstract

In this paper, I study the short-run impact of electrification on labor productivity in the U.S. during the period from 1910 to 1920, exploiting cross-county variation in electrical capacity. Using a fixed effects model, I estimate that a 1 percent increase in electric capacity leads to an approximate 12 percent increase in labor productivity, measured by total output per labor. Additionally, I find that electrification increased capital intensity, with a 1 percent increase in electric capacity resulting in a 10 percent rise in capital per labor. My estimates show no significant correlation between electrification and the capital-output ratio.

1 Introduction

I investigate whether electrification drives industrial development and, consequently, increases labor productivity. The belief that electrification is a prerequisite for industrial development was widely held among economists and policymakers. Based on

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this view, understanding the consequences of improving electricity access remains a perennial research topic. Whether electrification positively impacts labor productivity is unclear ex-ante. On one hand, electrification could enhance labor productivity by encouraging capital investment, which complements labor during the production process. On the other hand, electrification could be a costly investment with little causal impact. Thus, I aim to provide empirical evidence of the short-run causal effects of electrification on labor productivity in this paper.

Identifying the causal link between electrification and labor productivity is empirically challenging due to the inherent endogeneity of infrastructure investments such as electrification. Locations experiencing different levels of electrification might differ substantially in other ways. Additionally, electrification and industrial development often occur simultaneously in most economies. Large-scale randomized electrification is not feasible due to costs and practicality. These factors pose difficulties in identifying the causal effect of electrification. To control maximally for unobservable variables, geographical differences, and time trends in different locations, I implement a fixed-effects model that exploits county-level variation in electric capacity from 1910 to 1920. A key feature of this approach is that the fixed effects allow a flexible structure between unobservable confounding factors and the treatment variable, reducing the possibility of omitted variable bias. My identification relies on the assumption that, after controlling for observable factors in my dataset and all fixed effects, manufacturing industries in counties experiencing electrification extensions in 1910 would have trended similarly to their counterparts in counties with no such extensions. Under these assumptions, my estimates provide credible causal inference on the treatment effect of electrification.

For my empirical analysis, I use data from two sources. First, I use the Census of Manufactures from 1860-1940, compiled by [Lafortune et al. \(2019\)](#), to construct my outcome variables. My main outcome variable of interest, labor productivity, is defined as the difference between the total value of output and the total value of cost, divided by the number of workers. My electrification data are derived from power plant generation and location records found in the 1911 and 1919 editions of "Central Station Directory: A Complete List of Electric Light and Power Companies

with Data,” published by McGraw and digitized by [Vidart \(2024\)](#). Using this information, I construct measures of the electric capacity generated within each county in the U.S. from 1910 to 1920, a period when electrification was a major technological breakthrough. The sample used in my main estimation includes information on manufacturing outcomes at the county-by-industry level and measures of electrification at the county level for 118 counties, covering all major metropolitan areas at the time.

My estimated results suggest a substantial and immediate productivity gain from electrification. Specifically, I find that a 1 percent increase in electric capacity from 1910 to 1920 leads to an 8 percent increase in labor productivity on average. According to the current literature, the rapid productivity gain from electrification can be explained through multiple channels. First, electrification investment directly reduces the fixed costs of using energy. As documented by [Woolf \(1984\)](#), the lower price of electricity was a key driver of electrification in manufacturing sectors at the time. Second, electrification increases the complexity of the production process and the industry’s capital intensity. I find that a 1 percent increase in electric capacity leads to a 10 percent increase in capital per labor. The response of capital is consistent with the view that energy complements capital during production ([Goldin and Katz, 1998](#)). Lastly, new sources of power may increase the division of labor ([Atack et al., 2019](#)), which also leads to higher production efficiency.

This paper contributes to the literature empirically quantifying the effects of electrification during the early 20th century in the U.S., when electrification was a major breakthrough innovation ([Gray, 2013](#); [Kitchens, 2014](#); [Kitchens and Fishback, 2015](#); [Gaggl et al., 2021](#); [Lewis, 2018](#)). Most papers in this literature have focused on rural electrification, whereas my paper examines the effects of electrification in major metropolitan areas over a relatively short time span. Additionally, my paper relates to the literature explaining manufacturing productivity gains through technological innovation ([Crafts, 2002](#); [David and Wright, 1999](#); [Edquist and Henrekson, 2006](#); [Gordon, 2010](#); [Fiszbein et al., 2020](#)). This strand of literature indicates that productivity gains from technological innovation were sizable but slow, while my estimates indicate an immediate effect from electrification. With in this literature, my

paper is particularly related to the work of [Fiszbein et al. \(2020\)](#), which studies the impact of electricity on manufacturing industries. Fiszbein’s identification exploits the pre-electrification variation in energy intensity across industries combined with variation across locations in proximity to early hydroelectric power plants, while my paper directly exploits variation in electric capacity at the county level during 1910 and 1920.

The remainder of this paper is organized as follows. In section 2, I describe the data used in the empirical analysis. In section 3, I present the regression model and identification assumption. In section 4, I discuss the estimated results and robustness analysis. The final section concludes.

2 Description of Data

For my empirical study, I combine data from two sources. First, to construct the measure of the manufacturing outcomes, I utilize an industry-city-year level panel dataset built from the Census of Manufactures from 1860-1940 to construct the outcome variables. Second, I incorporate county-level electrification data for 1910 and 1920, digitized by [Vidart \(2024\)](#), to construct my treatment. The merged panel dataset contains information on total annual revenue, total cost of raw materials, total wages paid, total value of capital invested (including buildings and lands), total number of workers, and number of establishments at the industry-by-county level, as well as the measure of electric capacity at the county level, for the years 1910 and 1920. The sample used for the estimation of my main regression model includes 118 counties and 58 industries, resulting in a total of 4,213 observations.

The means and standard deviations of the relevant variables are summarized in [Table 1](#).

[[Table 1](#)]

2.1 Electricity data

The electrification data are derived from power plant generation and location records found in the 1911 and 1919 editions of "Central Station Directory: A Complete List of Electric Light and Power Companies with Data", published by McGraw and digitized by [Vidart \(2024\)](#). Each edition provides a comprehensive list of all electric power plants owned by electric and power companies in the United States, organized by city. Each city entry includes information on all the utilities operating within the city, detailing their generation capacity and other pertinent information. For a detailed description of the data construction, refer to the online appendix in [Vidart \(2024\)](#).

To construct my treatment variable, I use the total electric capacity within and 50 miles around each county's boundaries in 1911 and 1919, measured in 100s of megawatts¹, as my preferred treatment definition. Due to the constraints of energy transmission technology at the time, it was unfeasible to consume power far from the generation site before the expansion of high-voltage AC transmission lines in the late 1920s ([Delea and Casazza, 2011](#)). Therefore, I use this definition as an approximation of the extent of electrification in each county during the 1910s. To align with my outcome variables, I will use the electric capacity in 1911 and 1919 as proxies for the electric capacity in 1910 and 1920, respectively.

In [Figure 1](#), I present county-level maps illustrating the change in total electrical capacity within and 50 miles around county boundaries between 1910 and 1920.

[[Figure 1](#)]

2.2 Manufacture data

I utilize the decennial Census of Manufactures from 1890-1940, as compiled by [Lafortune et al. \(2019\)](#), to construct my variables for manufacturing outcome. Due to limitations in the availability of my electrification data, the analysis is concentrated on the period following the year 1910. From this dataset, I obtain measures for various

¹The 100-megawatts generating threshold is equivalent to 3 to 4 large generating plants.

outcomes at the industry-by-county level. My analysis centers on key variables such as capital, total output, number of workers, wage expenses, and number of establishments, all of which are pertinent to labor productivity. The geographic coverage of the Manufacturing Census varies by year. In 1910 and 1920, only cities with populations exceeding 50,000 were included. To align with the units in my electrification data, I have chosen to use counties as the unit of analysis. Accordingly, I matched each city to its corresponding county. This dataset encompasses approximately 60% of all manufacturing workers and manufacturing value added on average. The counties included in my sample represent the largest metropolitan areas of the period, including counties whose populations were five to six times larger than the average U.S. county. Figure 2 illustrates the counties included in my sample, based on the 1920 county boundaries. Industries are classified into 57 categories. For detailed information on industry classifications, refer to the data description section in [Fiszbein et al. \(2020\)](#). In the industry-by-county-by-year analysis, I exclude the "All other industries" category, as it is not comparable across years or regions.

[Figure 2]

3 Empirical Strategy

My main analysis employs a fixed effects model with a continuous treatment variable representing the intensity of electrification in a given county. This model exploits the variation in electrification capacity across counties from 1910 to 1920 to examine its impact on manufacturing outcomes at the industry-by-county level. Specifically, I estimate the average treatment effect of electrification across industries and counties by estimating the following model:

$$\begin{aligned} \log Outcome_{ict} = & \alpha + \beta \log Electrification_{ct} \\ & + X'_{ict} \delta + \mu_c + \mu_t + \mu_{it} + \gamma_t f(y_c) + \gamma_t f(x_c) + \epsilon_{ict}. \end{aligned} \quad (1)$$

I use μ_c , μ_t , and μ_{it} to denote county, year, and industry-by-year fixed effects, which absorb any factors that vary only at the county, year, and industry-by-year level,

respectively. X_{ict} stands for the control variables. Following the approach used by [Hornbeck and Rotemberg \(2024\)](#); [Donaldson and Hornbeck \(2016\)](#), I include a cubic polynomial in county latitude and longitude interacted with the year fixed effects ($\gamma_t f(y_c)$ and $\gamma_t f(x_c)$) to flexibly control for geographic differences among counties. $Electrification_{ct}$ is defined as the total electric capacity within and 50 miles around each county’s boundaries, and β reports its average impact on manufacturing outcomes $Outcome_{ict}$ across industries. I focus on three productivity variables in this study; namely, $Outcome_{ict}$ refers to total output per labor $\frac{Y}{L}$, capital per labor $\frac{K}{L}$, and capital output ratio $\frac{K}{Y}$ respectively. The interpretation of the coefficient β depends on the choice of the outcome variables $Outcome_{ict}$. For example, when $Outcome_{ict} = \frac{Y}{L}$ (total output per labor), β captures the average treatment effect of electric capacity on labor productivity across manufacturing industry sectors; when $Outcome_{ict} = \frac{K}{L}$ (capital per labor), β examines how industry’s overall capital intensity adjusts to the changes in electric capacity. This equation allows me to examine how labor productivity and capital intensity in various manufacturing industries respond to electrification.

Identification in this study relies on the assumption that, after controlling for observable factors in my dataset and all fixed effects, manufacturing industries in a county experiencing an electrification extension in 1910 would have trended similarly to their counterparts in counties with no such electrification extension. This assumption is essential to isolate the causal impact of electrification from other potential confounding influences².

A primary empirical concern in estimating the impacts of infrastructure investment is the potential endogeneity of infrastructure placement. Typically, infrastructure investments, such as electrification, are directed towards counties that may otherwise evolve differently over time ([Straub, 2011](#)). This selection bias implies that electrification investment might be endogenous due to the omitted variable problem, potentially leading to spurious correlations between increases in county electrification intensity and manufacturing outcomes. Although my specification absorbs variations

²[Callaway et al. \(2024\)](#) generalize the canonical parallel trends assumption under the classical DID setting into the case where the treatment is continuous.

at the county, year, and industry levels, and I incorporate controls for flexible geographical differences in my model, to further test for the pre-trend among different counties, I report the event-study estimation in the next section.

A second concern arises from recent literature discussing the decomposition of the treatment parameter in two-way fixed effects models (Roth et al., 2023; Goodman-Bacon, 2021; De Chaisemartin and d’Haultfoeuille, 2020). Essentially, the estimated treatment parameter in a fixed effects model can be decomposed into a weighted average of the treatment effects across different groups³. Under the presence of heterogeneous treatment effects, where the parameter β varies across groups, it is indeed possible that the weights assigned to each group’s treatment effect in the overall estimation are non-convex⁴. In certain instances, it is possible for the estimated overall treatment effect $\hat{\beta}$ to be positive, while the group-level treatment β_g is negative for all groups, due to the negative weights. To address this concern, I report the result of heterogeneous analysis in the next section as well.

4 Results

Table 2 presents results from estimating Equation 1. I estimate the electrification expansion has immediate and substantial impacts on labor productivity. Column reports that a 1 percent increase in electric capacity from year 1910 to 1920 leads to an approximate 12 percent increase in labor productivity (Panel A), calculated as (total revenue minus total costs) divided by the number of workers, in my baseline model. When accounting for geographical differences and average factory size of the industry, the estimates decrease to about 8 percent. Nonetheless, the results remain statistically significant at the 99% confidence level. Along with the increase in labor productivity, I also observe a significant rise in capital intensity, measured as capital per labor. Panel B reports that a 1 percent increase in electric capacity leads to an approximately 10 percent increase in capital per labor. However, when controlling for geographical differences and factory size, the results become less significant. In

³In my analysis, these groups are defined by different industries located within various counties

⁴Non-convex means negative weights, i.e, $\beta = \sum_g w_g \beta_g$, where $w_g < 0$ for some group g .

contrast to the output-labor ratio, the estimates suggest that there is no significant relationship between electrification and the capital-output ratio (Panel C). Lastly, I find that electrification has a significant impact on labor expenditure per worker and total material expenditure, where a 1 percent increase in electric capacity leads to a 21 percent increase in labor expenditure per worker and a 25 percent increase in total material expenditure.

My findings are consistent with the intuition that higher electric capacity provides access to a cheaper source of energy, which leads to industry capital deepening due to the complementarities between electricity and capital (Goldin and Katz, 1998). This conclusion is supported by the positive effect of electric capacity on capital per labor. The productivity growth observed between 1910 and 1920, as presented in my regression results, was likely driven by increased capital investment due to lower energy costs. On the other hand, my findings indicate that this productivity growth is unlikely to be attributed to changes in the production process, as there is no statistically significant effect between electricity and the capital-output ratio. Additionally, as a byproduct of labor productivity growth, I find that labor expenditure per worker increased with the rise in electric capacity.

[Table 2]

4.1 Dynamic specification

Recall that the main identification assumption I invoke in my primary specification is that industry sectors in counties experiencing large electrification extensions in 1910 would have trended similarly to their counterparts in counties experiencing smaller electrification extensions. In this section, I present "event study" estimates of the effect of electrification to verify this assumption. Consider the following equation:

$$\begin{aligned} \log Outcome_{ict} = & \sum_{t \in \{1860, \dots, 1900\}} \gamma_t \cdot \Delta Ele_c \cdot 1\{year = t\} \\ & + \sum_{t \in \{1920, \dots, 1940\}} \beta_t \cdot \Delta Ele_c \cdot 1\{year = t\} + \mu_c + \mu_t + \mu_i + \epsilon_{ict}. \end{aligned} \quad (2)$$

This model is similar to Equation 1, except that I replace the original treatment variable with $\Delta Ele_c \cdot 1\{year = t\}$, where ΔEle_c is defined as the change in electric capacity at county c from year 1910 to 1920, and $1\{year = t\}$ is an indicator function for year t , for $t \in \{1860, 1870, 1880, 1890, 1900, 1920, 1930, 1940\}$. The fixed effects μ_i , μ_t and μ_c are included as defined previously. Notice that I excluded year 1910 to avoid multi-collinearity, this implies that all treatment estimates are relative to the year 1910. For $t > 1910$, β_t captures the dynamic treatment effects of the electricity investment on manufacturing outcomes. For $t < 1910$, γ_t captures any potential violation of the pre-trends assumptions which could threaten my identification strategy⁵. Using this equation, I examine whether my dependent variables of interest exhibit differential trends before the arrival of electrification extension.

The estimated results for Equation 2 are reported in Table 3. Panel A in Figure 3 summarizes the dynamic effects of changes in electrical capacity from 1910 to 1920 on labor productivity, relative to the year 1910. The figure shows $\Delta Ele_c \cdot 1\{year = t\}$ coefficients along with 95 percent confidence intervals. In Figure 3, there is no evidence of any differential trend in labor productivity, before the year 1900. Indeed, all the coefficients prior to year 1900 are statistically insignificant. However, the coefficient for $\Delta Ele_c \cdot 1\{year = 1900\}$ is statistically significant, suggesting that potential pre-trends may exist. These estimates might invalidate the causal interpretation of Equation 1.

For completeness, I report the estimates for the same model using capital per labor as the outcome variable in Panel B. The results are more noisy in this setting. Note that the years 1930 and 1940 are excluded since I do not have data on total capital for those years. Lastly, the results using the output-capital ratio as the outcome variable are excluded, as the treatment effect of electric capacity on it is insignificant.

[Figure 3]

⁵Roth (2022); Sun and Abraham (2021) discuss the limitation of using event-study design as a test for pre-trends.

4.2 Heterogeneous analysis

Thus far, my estimates suggest that an increase in electric capacity has a positive effect on labor productivity across industries on average. However, there may be heterogeneous treatment effects, driven at least in part by the fact that industries requiring higher levels of energy intensity could be more susceptible to changes in electric capacity. In this section, I explore the role of heterogeneity in the response of labor productivity to changes in electric capacity. Specifically, I categorize industry sectors into four groups based on the quartile of their energy usage intensity, measured by horsepower usage per output using 1910 data. For instance, industries such as dairy products, leather products, and tobacco manufacturing are considered less energy-intensive and fall into the first quartile of energy usage in 1910. On the other hand, heavy industries such as iron and steel manufacturing are in the fourth quartile based on their energy usage in 1910. Based on this classification, I capture the extent of heterogeneity in the treatment effects among different industry sectors by estimating Equation 1 separately by the quartile of the industry energy intensity. The most ideal scenario would be that industries with different energy intensities are equally affected by changes in electric capacity, thereby eliminating concerns about the "negative weights problem" discussed in recent fixed-effects literature. However, simple economic reasoning suggests that the absence of heterogeneity is unlikely in this context.

Panel A in Table 4 presents the estimated coefficients of electric capacity using labor productivity as the dependent variable. These results show limited evidence of heterogeneity across industries: the coefficients for electric capacity range from 0.07 to 0.26, compared to 0.12 in the baseline model. In general, I find that electric capacity has positive treatment effects on labor productivity on average, regardless of the level of energy usage in an industry. To evaluate whether the variation in treatment effects is substantial enough to cause any issues discussed in recent fixed-effects literature, precise weights of the average treatment effect need to be calculated using the method discussed in [De Chaisemartin and d'Haultfoeuille \(2020\)](#), which is beyond the scope of this study. Surprisingly, the magnitude of the treatment

effect shows no clear pattern with respect to the level of industry energy intensity. According to the estimates, industry sectors in the 2nd quartile receive the greatest impact from the change in electric capacity, while industry sectors in the 3rd quartile experience the smallest impact. Industries in the 1st and 4th quartiles fall in between.

Panel B and Panel C present the estimates using capital per labor and capital-output ratio, respectively. The treatment effect of electrification is substantially heterogeneous across industries for capital per labor, with estimates ranging from 0 to 0.19. Notably, only the coefficient for the 2nd quartile is significantly different from zero. The coefficients for the capital-output ratio are insignificant across all energy intensity levels, as expected.

[Table 4]

5 Conclusion

In this paper, I study the impact of electrification on labor productivity in the U.S. during the period from 1910 to 1920, exploiting cross-county variation in electrical capacity. Using a fixed effects model, I estimate that a 1 percent increase in electrical capacity leads to an approximate 12 percent increase in labor productivity, measured by total output per labor. Additionally, I find that electrification increased capital intensity, with a 1 percent increase in electrical capacity resulting in a 10 percent rise in capital per labor. My estimates show no significant correlation between electrification and the capital-output ratio. These findings underscore the critical role of infrastructure investment in manufacturing industries, offering a relevant lesson for many developing countries today.

In the context of my future research directions, I address several limitations within this study. First, while other studies have suggested that rapid productivity gains may be linked to access to cheaper electricity, the precise underlying mechanism remains unexplored in my investigation. A more sophisticated economic model is needed for further understanding. For example, [Fried and Lagakos \(2023\)](#) explain the productivity gain through a dynamic model. Second, my data on manufacturing

outcomes only covers counties with populations greater than 50,000 during the period from 1910 to 1920, amounting to 125 counties. Expanding the dataset to include all counties in the U.S. would significantly improve the precision of the estimations. Lastly, recent advances in the difference-in-differences (DID) literature suggest that results from classical fixed effects models should be interpreted with caution ([Roth et al., 2023](#)). Applying more sophisticated econometric estimators that are robust to violations of traditional assumptions in DID models could enhance the robustness of the estimates.

A Figures and Tables

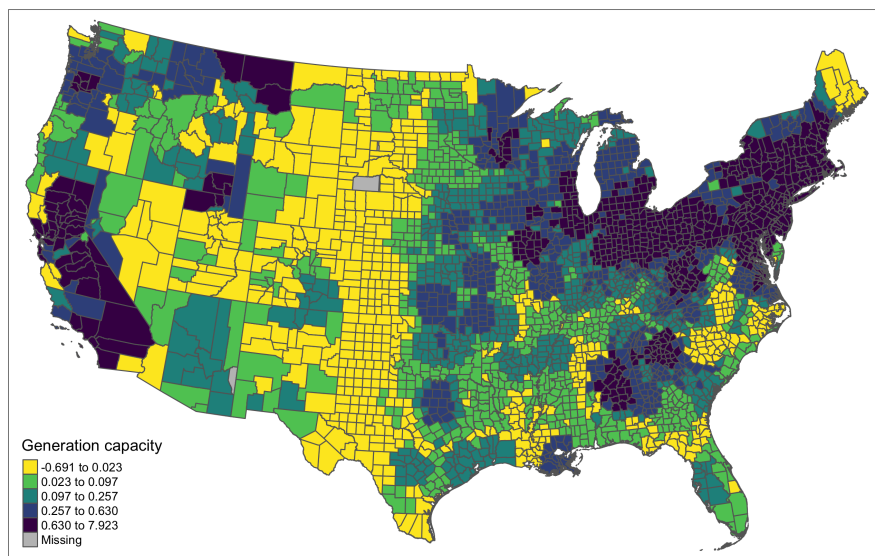


Figure 1: Change in Electrical Capacity from 1910 to 1920

Notes: This figure present the change in electrical capacity in U.S. from 1910 to 1920.

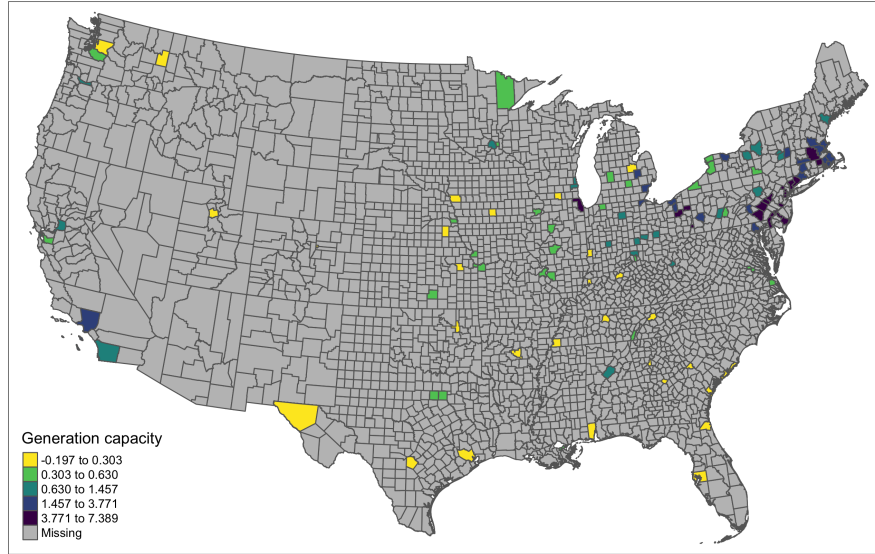


Figure 2: Counties included in the Dataset

Notes: This figure present the counties that enter my sample.

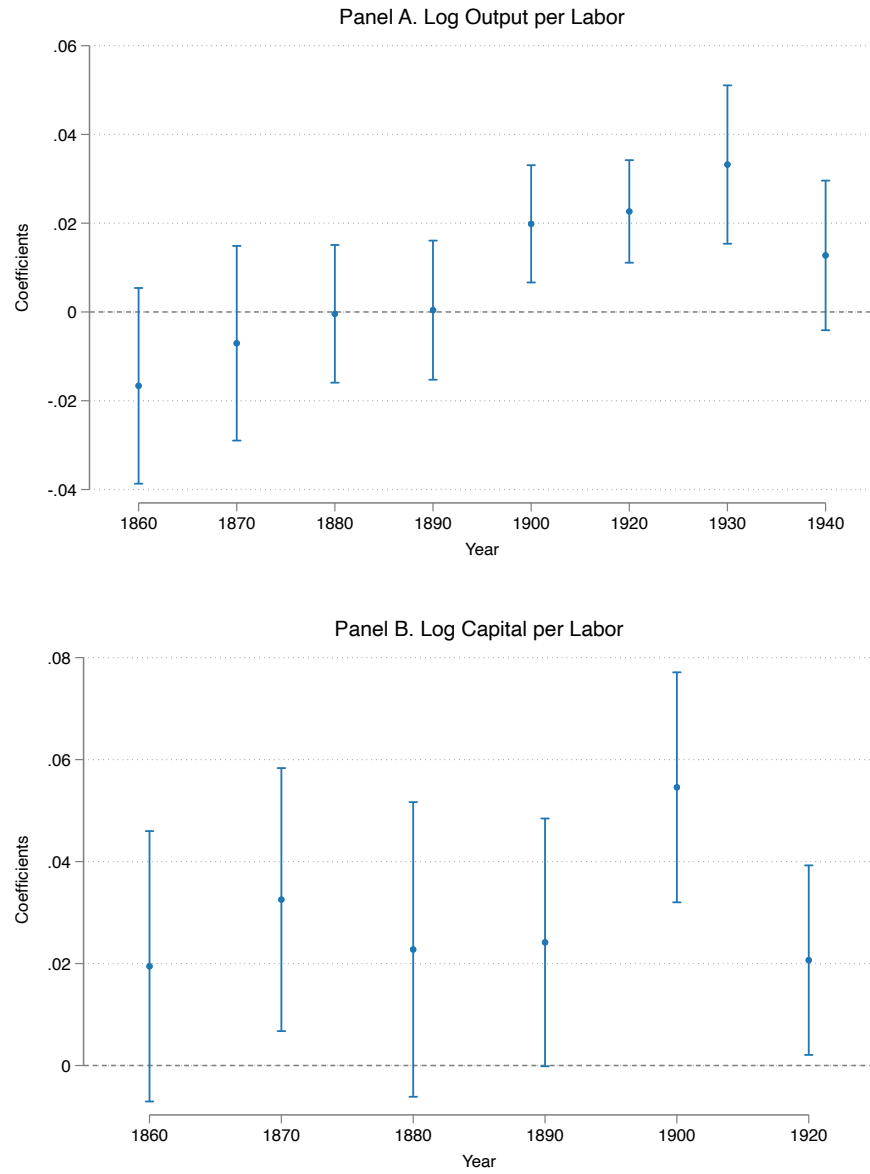


Figure 3: Event-study plot

Notes: This figure plots the coefficients of Equation 2. The outcome variables in panel A and panel B are log total output per labor and log capital per labor, respectively. The year 1910 is used as baseline.

Table 1: Summary Statistics

	(1)	(2)
	1910	1920
Electrification capacity	1.218 (1.241)	2.913 (2.795)
Value added	2227534.164 (9776812.585)	5256116.093 (46475159.119)
Total revenue	4806424.347 (20819664.935)	12061256.567 (1.029e+08)
Capital	3608797.816 (12049621.764)	6860693.731 (39457300.838)
Number of workers	1688.453 (7105.198)	2023.353 (13913.971)
Labor expenditure	802076.726 (3492942.256)	1994231.888 (15871588.795)
Material expenditure	2683767.559 (12884947.108)	6804709.694 (58243104.031)
Number of establishments	51.826 (246.901)	60.123 (593.625)
Observations	1589	2624

Note: Standard deviations are in parentheses.

Table 2: The Effect of Electrification on Manufacturing Outcomes

	Baseline specification (1)	Add geo difference (2)	Add establishment size (3)
<i>Panel A. Log Output per Labor</i>			
Log Electrification	0.120*** (0.023)	0.073*** (0.022)	0.080*** (0.022)
Observations	4,185	4,185	4,185
R^2	0.620	0.621	0.628
<i>Panel B. Log Capital per Labor</i>			
Log Electrification	0.100*** (0.035)	0.068* (0.038)	0.059 (0.038)
Observations	4,202	4,202	4,202
R^2	0.591	0.592	0.598
<i>Panel C. Log Capital Output ratio</i>			
Log Electrification	-0.014 (0.033)	0.000 (0.039)	-0.016 (0.039)
Observations	4,189	4,189	4,185
R^2	0.364	0.365	0.393
<i>Panel D. Log Labor Expenditure</i>			
Log Electrification	0.213*** (0.072)	0.151** (0.058)	0.155*** (0.058)
Observations	4,200	4,200	4,200
R^2	0.700	0.704	0.706
<i>Panel E. Log Material Expenditure</i>			
Log Electrification	0.257** (0.113)	0.252** (0.124)	0.088 (0.088)
Observations	4,206	4,206	4,202
R^2	0.582	0.585	0.833

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the county level. The baseline specification controls for county fixed effects, year fixed effects, and industry-by-year fixed effects. The second column further includes year-interacted cubic polynomials in county latitude and longitude to control for geographical differences. The third column adds Log establishment size in the regressions.

The dependent variables are: $\log \frac{TotalOutput}{NumberOfWorkers}$, $\log \frac{TotalCapital}{NumberOfWorkers}$, $\log \frac{Capital}{TotalOutput}$, $\log \frac{TotalLaborExpenditure}{NumberOfWorkers}$, and $\log TotalMaterialExpenditure$.

Table 3: Event Study

	Log Output per Labor (1)	Log Capital per Labor (2)
chele1860	-0.017 (0.011)	0.019 (0.013)
chele1870	-0.007 (0.011)	0.033** (0.013)
chele1880	-0.000 (0.008)	0.023 (0.015)
chele1890	0.000 (0.008)	0.024* (0.012)
chele1900	0.020*** (0.007)	0.055*** (0.011)
chele1920	0.023*** (0.006)	0.021** (0.009)
chele1930	0.033*** (0.009)	
chele1940	0.013 (0.009)	
Observations	20,273	17,780
R^2	0.699	0.641

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the county level. The baseline specification controls for county fixed effects, year fixed effects, industry fixed effects, and industry-by-year fixed effects. The second column further includes year-interacted cubic polynomials in county latitude and longitude to control for geographical differences. The third column adds Log establishment size in the regressions.

Table 4: Heterogeneous Analysis

	1st quartile (1)	2nd quartile (2)	3rd quartile (3)	4th quartile (4)
<i>Panel A. Log Output per Labor</i>				
Log Electrification	0.150** (0.067)	0.265** (0.101)	0.074* (0.043)	0.103** (0.049)
Observations	644	634	1,500	1,374
R^2	0.551	0.462	0.491	0.344
<i>Panel B. Log Capital per Labor</i>				
Log Electrification	0.137 (0.095)	0.191* (0.111)	-0.002 (0.061)	0.082 (0.075)
Observations	644	635	1,507	1,383
R^2	0.366	0.385	0.279	0.202
<i>Panel C. Log Capital Output Ratio</i>				
Log Electrification	-0.014 (0.097)	-0.069 (0.126)	-0.066 (0.062)	-0.019 (0.062)
Observations	644	634	1,500	1,374
R^2	0.189	0.341	0.170	0.266

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors are clustered at the county level. The baseline specification controls for log firm size, county fixed effects and year fixed effects. The second column further includes year-interacted cubic polynomials in county latitude and longitude to control for geographical differences. The third column adds Log establishment size in the regressions.

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